LC9203DB

Li Battery Transfer to Dry Battery

GENERAL DESCRIPTION

LC9203DB is a SOC that it integrate Li-battery charger, Buck discharge ,and backflow PMOS. It has perfect charge and discharge management and protection functions of Li-battery。

LC9203DB can output 1.5Vout 2A with LiBattery Input and also can charge with 500mA current.

LC9203DB has lower quiescent current less 8uA.

It only need few components and can reduce the BOM area and BOM cost.

FEATURES

- Integrated charging and 1.5V discharge management
  LC9203DB-3.7V Support 3.7V Li-Battery
  LC9203DB-3.2V Support 3.2V Li-Battery
- Output Voltage 1.5V
- 1.5MHz Constant Frequency Operation
- Charging current up to 0.5A
- 2A Ouput Capability
- Low Quiescent Current 6uA
- High Efficiency up to 92%
- Adjust output voltage from 1.5V to 1.1V with battery voltage
- Under Voltage Lockout, Over Current, and Short circuit Protection
- Preset 4.2V Charge Voltage with ±1% Accuracy
- 2.7V Trickle Charge Threshold
- DFN3*3 8L Package
- RoHS Compliant and 100% Lead(Pb)-Free

TYPICAL APPLICATION

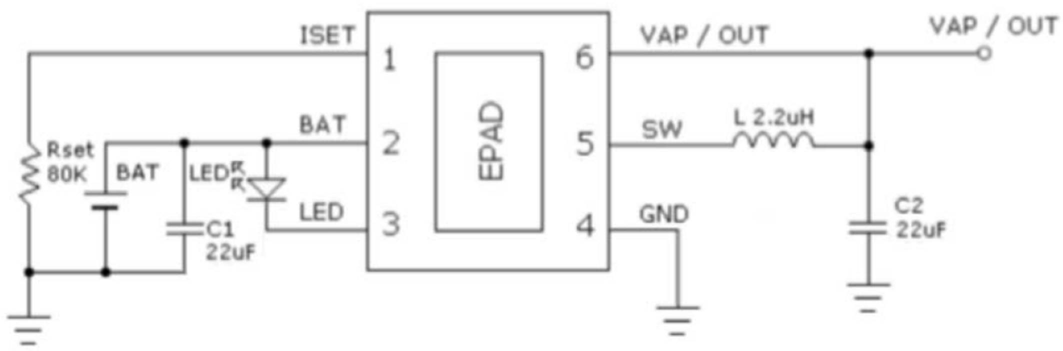
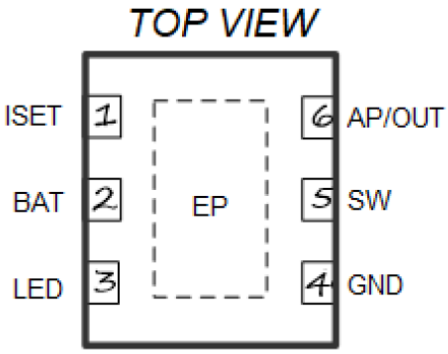


Fig1 : Typical Application

PIN CONFIGURATION



2mmX2mm 6-Pin DFN Package

Pin No.	Pin	DESCRIPTION
1	I_{SET}	Charging current setting terminal
2	BAT	Charging current output terminal
3	LED	Charging state indicating terminal
4	GND	Ground
5	SW	Output pin of MOS, connected with inductor
6	V_{AP}/OUT	Adaptor input voltage pin and 1.5V battery output pin

ABSOLUTE MAXIMUM RATINGS

(Note: Do not exceed these limits to prevent damage to the device. Exposure to absolute maximum rating conditions for long periods may affect device reliability.)

Symbol	Parameter	Min.	Max.	Unit
V_{PIN}	Pin Breakdown Voltage	-0.3	7	V
T_{OP}	Operating Temperature	-40	85	℃
T_{STG}	Storage temperature	-55	150	℃
ESD	Human being mode（HBM）	--	2000	V
	Mechanism mode（MM）	--	200	V
V_{AP}	Adaptor voltage	4.5	5.5	V
I_{CHG}	Charging Current	--	500	mA
I_{OUT}	Discharging Current	--	2	A

ELECTRICAL CHARACTERISTICS (V_{IN}=5V，Ta=25℃,if not otherwise noted.)

Parameter	Symbol	Testing conditions	Min.	Typ.	Max.	unit
Charging						
Adaptor Input Voltage	V_{AP}		4.7	5.2	5.7	V
	OVP	R_{SET} = 69.8K		6.5	7.0	V
Regulated Output Voltage	V_{CV}	**3.7V**, current drops down to I_{CHG} /10	4.15	4.20	4.25	V
		3.2V, current drops down to I_{CHG} /10	3.66	3.70	3.74	V
Recharge Voltage	V_{RCHG}	3.7V Battery	3.99	4.03	4.07	V
		3.2V Battery	3.51	3.55	3.59	V
Charging Current	I_{CHG}	R_{SET} = 69.8K	450	500	550	mA
Trickle Charging Current	I_{TRIKL}	V_{BAT}< V_{TRIKL} , R_{SET} = 69.8K	40	50	60	mA
Trickle Charging Threshold Voltage	V_{TRIKL}	R_{SET} = 69.8K, V_{BAT} rise, **3.7V**	2.6	2.7	2.8	V
		R_{SET} = 69.8K, V_{BAT} rise, **3.2V**	2.3	2.4	2.5	V
Trickle Charging Hysteresis Voltage	V_{TRHYS}	R_{SET} = 69.8K, **3.7V**	150	200	250	mV
		R_{SET} = 69.8K, **3.2V**	100	130	160	mV
V_{CC} - V_{BAT} Threshold Voltage	V_{ASD}	V_{CC} rise	60	100	140	mV
		V_{CC} falling	60	100	140	mV
ISET Pin Voltage	V_{SET}	R_{SET} = 69.8K	0.95	1.0	1.05	V
Charging OTP Threshold	T_{OTP1}			130		℃
Discharging						
Quiescent Current	I_Q	Empty load（Buck mode）		6	10	uA
	I_{SD}	UVLO（shut down mode）		2	3	uA
Output Voltage	V_{OUT}	BUCK mode	1.48		1.54	V
Oscillator Frequency	f_{SW}	Iout= 500mA		1.5		MHz
Low Battery Output Voltage	V_{OUT}	BUCKmode	1.05	1.1	1.15	V
Low Battery Voltage	V_{BAT_LOW}	3.7V Battery	3.10	3.15	3.20	V
		3.2V Battery	2.40	2.45	2.50	V
Low Battery Hysteresis Voltage	V_{BAT_LOWHYS}	3.7V Battery		150	250	mV
		3.2V Battery		100	150	mV
UVLO Threshold	UVLO	3.7V Battery	2.70	2.75	2.8	V
		3.2V Battery	2.05	2.10	2.15	V
UVLO Hysteresis Voltage	UVLO_hys	3.7V Battery	200	250	300	mV
		3.2V Battery	140	170	200	mV
Output Current	Iout			2		A
Peak Inductor Current1	I_{LIMT1}		3.5			A
Peak Inductor Current2	I_{LIMT2}	V_{OUT} <0.4V	2.5			A
OTP Threshold	T_{OTP2}			150		℃
OTP Hysteresis	$T_{OTP-HYS2}$			20		℃
LED Current	LED		4	5	6	mA

FUNCTION DIAGRAM

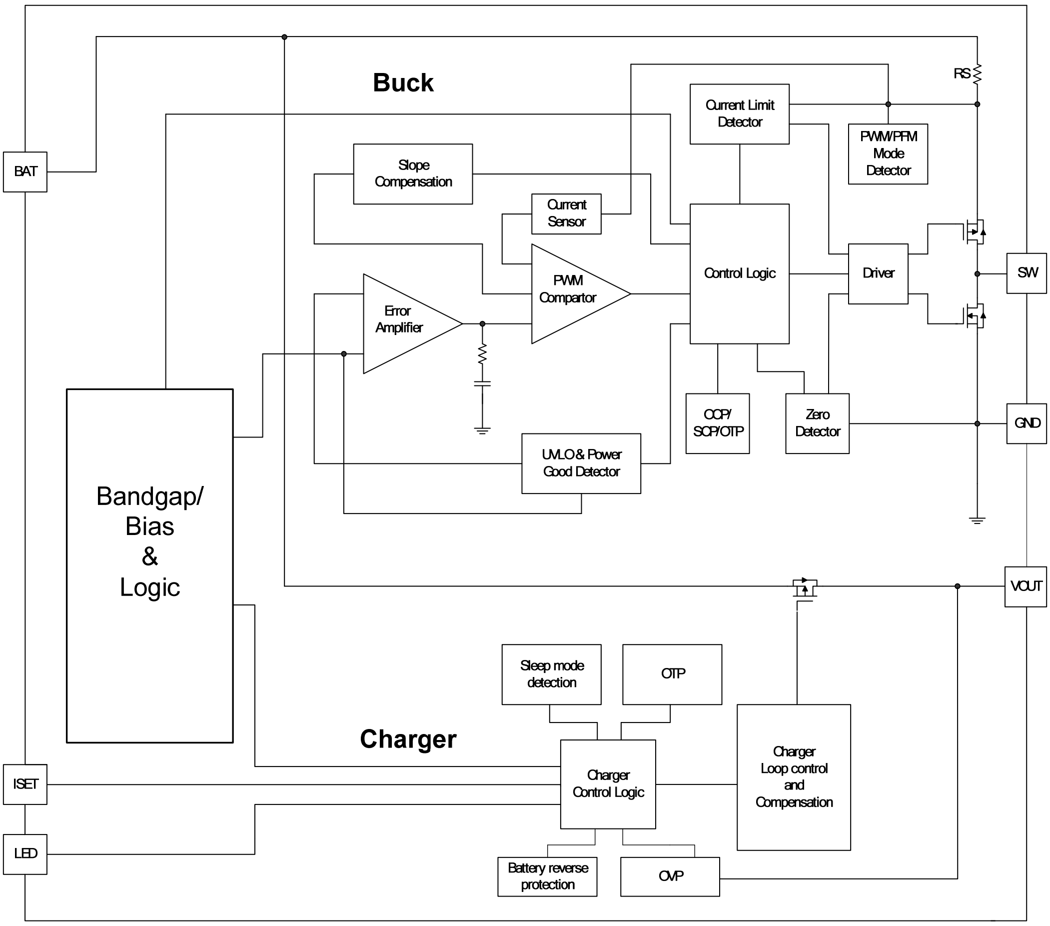
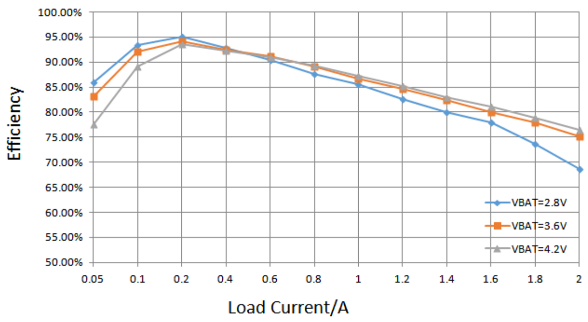


Fig2：Function Diagram

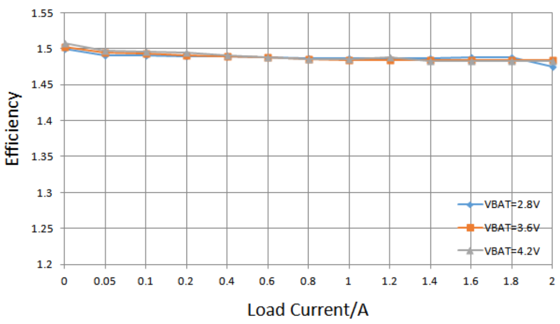
TYPICAL PERFORMANCE CHARACTERISTICS

(VIN=5V，V_{BAT}=3.6V，L1=2.2μH，C_B=22μF，C_L=22μF, if not otherwise noted.)

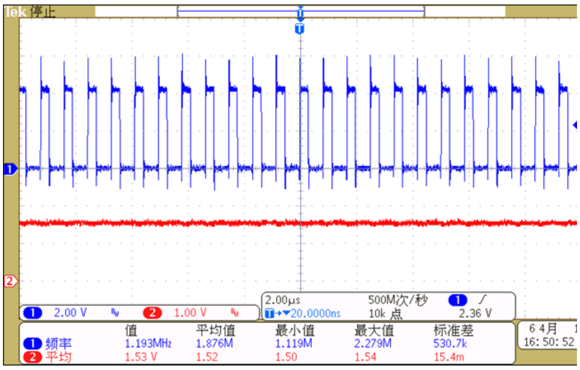
Efficiencv vs Load Current



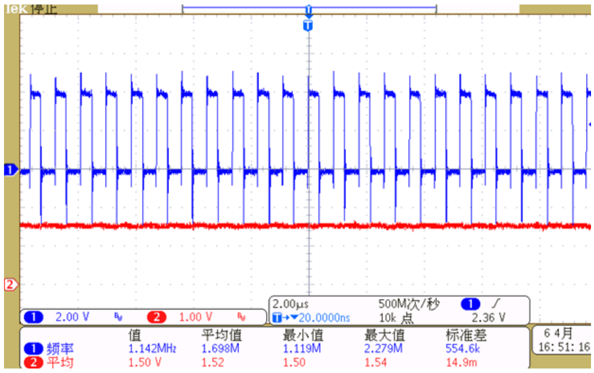
VOUT vs Load Current



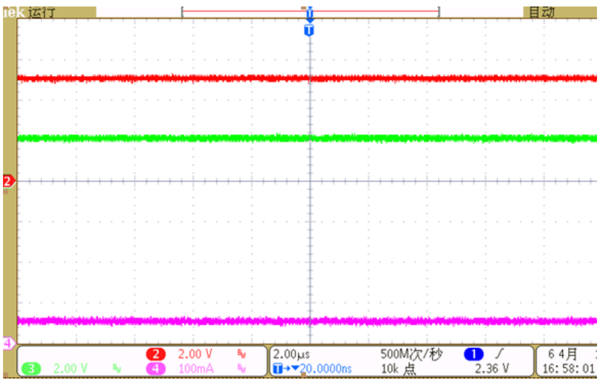
Iout=0.5A



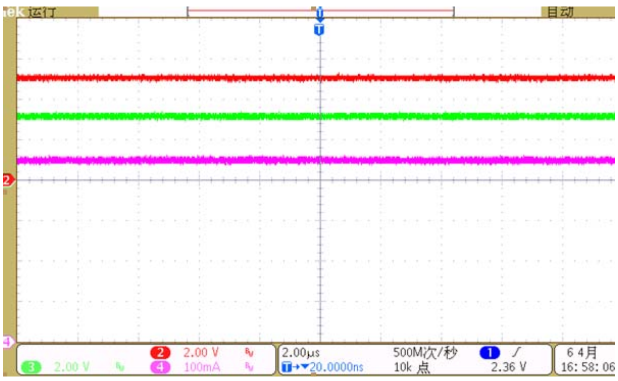
Iout=2A



Trickle charging



Constant current charging



FUNCTIONAL DESCRIPTION

Judgment of operating mode

Compare the voltage from VOUT with VBAT to judge the operating mode of the system.

When V_{OUT}>V_{BAT}+100m, LC9203DB works in Linear Charge mode.The OUT terminal acts as the power supply to charge the single Li-Baettery at the BAT terminal.

When V_{OUT}<V_{BAT}, LC9203DB works in Discharge mode. LC9203DB converts the voltage of Li- Battery to 1.5V (as dry battery voltage) and outputs from OUT terminal. When the battery voltage is lower than V_{BAT_LOW} The output voltage is reduced to 1.1V at low. The discharge mode uses very low power consumption mode under light load.Under light load, the discharge mode adopts very low power consumption mode, and the no-load quiescent current is only 6uA.

Discharging mode

LC9203DB achieves voltage reduction through external inductance (L1), output capacitance (C2) and switching on/off internal P-MOS main and N-MOS synchronous rectifier.

In normal operation, LC9203DB work in PWM mode and operating frequency remains constant. The chip is not any external components demanded due to an internal current-mode compensating circuit applied; LC9203DB provides the feed-forward circuit to enhance the performance of transient voltage reacting. P-MOS be opened along the descending edge of the sawtooth wave generated by the internal waveform generator. P-MOS be closed N-MOS be opened as PWM comparator overturning or voltage protecting and current limiting conditions changed. N-MOS be shut off as P-MOS be restarted or reverted current be detected.

LC9203DB will enter the PFM mode while the reverse current be detected and the maximum current of inductor are below 100mA, so as to decrease the operating current.

1）Over current protecting(OCP)

LC9203DB works in DC/DC mode, the over current protecting circuit internal is detecting the current through P-MOS. The P-MOS be closed as the current is exceeding the limiting threshold be turned off, refuse the current of the inductor to be added. In the next pulse cycle, the chip will return to normal operation while the current of P-MOS is less than the current limiting threshold, chip recoveries to normal operation from over current protecting. In case of over current occurring, P-MOS must be shut off and system enters over current protecting state.

2）Short circuit protecting

While VOUT terminal connected to ground due to short circuit, LC9203DB enters frequency decreasing mode. The system decreases the operating frequency, in order to reduce the input current of terminal VOUT, and cut down the heating of the circuit effectively.

The system return to normal DC/DC mode as the short circuit removed.

3）Synchronous rectifier

LC9203DB offers an N-MOS synchronous rectifier internal, thus no schottky rectifier diode external required. The turn-on voltage drop of LC9203DB is lower than the external schottky rectifier diode, therefore circuit efficiency be improved.

4）Over temperature protection(OTP)

The circuit turns P-MOS and N-MOS off as internal temperature of chip exceeded the threshold of OTP, the voltage output be forbidden. When operating temperature is dropped down and under OTP recovery threshold, the circuit renews to normal operating state, and P-MOS open automatically in next cycle.

Charging operating mode

LC9203DB starts a charging cycle as V_{OUT}>V_{BAT}+100mV.

The battery enters trickle charging state as V_{BAT}< V_{TRIKL}, under the concerned state,the cell charging current equals to 10% of the setting I_{CHG}, system starts safe treatment in advance.

The battery enters trickle charging state, while trickle charging current can make battery's voltage arising and keeps over V_{TRIKL}; this means battery's voltage be improved up to safe level and serves full-current charging.

As trickle charging ended, the adaptor enters mode of constant current charging and provides a setting charging current to the battery. As V_{BAT} is reaching to V_{CV}, LC9203DB enters mode of constant voltage charging. While the charging current is lowing to 10% of the setting charging current, IC's VLED terminal give a signal of charging completed, this remarks as battery full; but the circuit is charge in a weak current. The course may regard as a charging cycle been ended.

1）Charging current setting

Charging current can be set by an external resistor which connected between pin of PROG and the ground. The values calculates for current and resistor with formula listed following.I_{CHG} = (V_{SET} /R_{PROG})×35000

RPROG（K）	200K	150K	120K	100K	69.8K
IBAT（mA）	175mA	233mA	292mA	350mA	500mA

2）Charging completed

While BAT reaches to V_{CV}, a charging cycle be judged a charging completed as the charging current is lowing to 10% of the setting charging current.

3）Over temperature protecting（OTP1）

In the charging mode, when the internal temperature of LC9203DB exceeds the OTP threshold, the IC shuts off the loop of charging circuit to drop circuit's power down and protect chip not to be damaged; the circuit returns to normal charging state as chip's operating temperature lowing to the value of TOTP-TOTP-HYS.

LED lights indicating state

LC9203DB applies an open drain terminal for LED controlling, a LED light external connects a high-level voltage from LED pin, LED light works in a constant current of 3mA. Circuit LED pin's controls outputs as: high impedance/low-level/square wave, the outer light indicates different manners according to different charging states, details as:

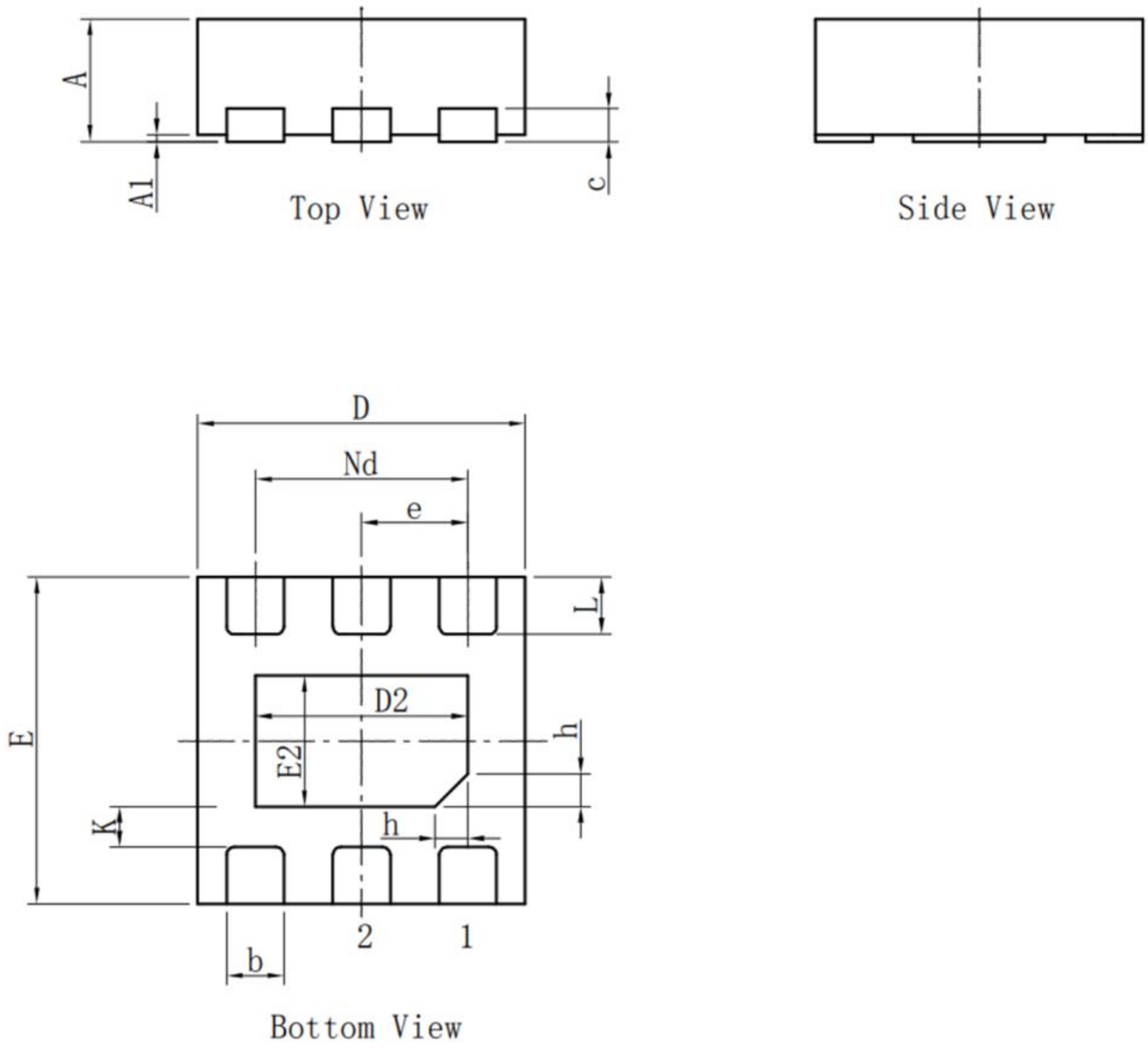
State	Charging	Fully	Discharging	Discharging short circuit	Discharge under voltage
LED light	1Hz flicker	lighting	lighting off	10Hz flicking rapidly	Hz flicking 8 times，off

Reminders for PCB design:

1, Meet return circuit in big current, lines of BAT/GND and etc. lead as wide as possible. 2, Capacity of BAT terminal closes to the chip possibly.
3, Inductance (L) closes to SW pin nearly.
4, Feed-back resister closes to chip set, GNDs of feed-back resister and chip close nearly.
5, GND setting makes more holes to connect bottom layer and lower return circuit's impedance. 6, GND line connects to battery, line keeps short as possible then 3cm long should be best.

PACKAGE DIMENSIONS(OUTLINE)

DFN2X2 6L



尺寸 / 标注	最小(mm)	标准(mm)	最大(mm)	尺寸 / 标注	最小(mm)	标准(mm)	最大(mm)
A	0.70	0.75	0.80	E2	0.75	0.80	0.85
A1	0.00	0.02	0.05	e		0.650BSC	
b	0.30	0.35	0.40	Nd		1.300BSC	
c	0.18	0.20	0.25	K	0.20	–	–
D	1.95	2.00	2.05	L	0.28	0.33	0.38
D2	1.25	1.30	1.35	h	0.15	0.20	0.25
E	1.95	2.00	2.05				